Introduction to Finite Element Analysis

Observation Lesson

Late Fall 2026

Overview

Previously, students have analyzed a frame under the weight of an engine using Autodesk Inventor.

Student Outcomes

- Create several sample finite element meshes.
- Create a FEA model in Python or a second model in Autodesk Inventor

Next Generation Science Standards

HS.ETS1.4

Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

Differentiation

Students have the choice of using Autodesk Inventor or coding in Python.

Materials

- Projector & Promethean board
- PowerPoint Slides
- Crash simulation video
- Autodesk Inventor
- Python
- Calculators
- Laptop

Assessment

Students will be assessed as they participate and share out in class, and complete mesh worksheets, and a project.

Lesson Agenda

Sample crash analysis video

RET Research PowerPoint

Discuss basics of FEA

Practice creating meshes

Discuss labeling mesh with nodes

Choose Inventor or Python project

Outline

- Students have analyzed a frame model using Autodesk Inventor.
- A sample video is shown to introduce quadrilateral finite element mesh and how FEA is used to simulate crash deformation in cars.
- A triangular mesh is introduced when RET summer research is summarized with a PowerPoint presentation.
- Summarize the finite element model basics: nodes, elements, loads, boundary conditions.
- Students practice creating a mesh by hand with grid paper with dots. They explore a basic square grid pattern, using the same dot pattern to create triangular mesh, and then use a second dot pattern to create an equilateral triangle mesh.
- Discuss how nodal numbering, node coordinates, and element connectivity are the starting points of creating your own model in Python.
- Students choose to either work in Autodesk Inventor or code in Python.
- Those who code in Python may use the "getfem" website and choose from sample code given in "Introduction to FEM Analysis with Python" webpage.

Pre-assessment

- Autodesk Inventor Frame Design of Engine Mount project
- Coding experience survey

Post Assessment

- Mesh practice worksheets
- Second Autodesk Inventor Analysis or Python code